Records, Vitals and Band — QA Test Cases

Task 5.10 · v0.4 · Owner: Syed Sarim Jawad Zaidi · From docs/api/records_contract.md, appointment_contract.md, vitals_contract.md and wearable_contract.md (Tasks 5.1–5.4, dev up to PR #60) · 5.9 mockups v0.2 / v0.3

Purpose: prove that records can be added, viewed and exported by the right people only, that appointments remind both sides, that vitals are recorded and shown correctly, and that the band adds data without ever blocking the app. Results go on a copy of the QA Checklist template.

Status: all four sections now use the real limits and error codes of Tasks 5.1–5.4. Section D runs on mock band data (POST …/wearable/mock); the real-band cases (W-09, W-10) wait for Task 5.5. v1.0 follows after the first run against Talha's screens.

1. Test setup

1.1 Test accounts

Role	Name in tests	Set-up
Dependent	Ammi	Registered, KYC verified, linked to Ali.
Primary Guardian	Ali	Phone + CNIC verified.
Secondary A	Sara	Invited by Ali, verified. Permission Health Records = ON.
Secondary B	Bilal	Invited by Ali, verified. Health Records OFF.
Outsider	Zain	A verified Guardian NOT linked to Ammi (for refusal checks).

1.2 How to run

- **Order:** backend checks through Postman as soon as Abdullah's 5.1–5.4 are merged; app checks after Talha's 5.6–5.8.
- **Test files:** one small PDF lab report, one phone photo of a prescription, one X-ray image, one non-medical file (e.g. .exe or .zip) and one very large file.
- **Band:** use mocked band data first (roadmap 5.4); repeat section D with the real band after 5.5.

2. Test cases

A. Medical vault (FR-G1, FR-G2, FR-G3)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
R-01	FR-G1	Ali logged in.	Medical Vault → Add record → Lab report, title "HbA1c", date, upload a PDF → Save.	201; record in the list (15) with category, record date and "added by Ali"; filter Lab shows it; file icon shown.		
R-02	FR-G1	Ammi logged in.	My health → My records → add a prescription photo with the camera.	Saved with "added by Ammi"; Ali sees it too.		
R-03	FR-G1	—	Add a record with only a note (no file); then one with neither file nor note.	Note-only record saved (no file icon; opening its file gives 404 NO_FILE). Empty one: 422 RECORD_NEEDS_CONTENT.		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
R-04	FR-A4	Sara has health_records ON, Bilal OFF.	Both open Ammi's vault; then Ali switches Sara OFF and she refreshes.	Sara sees the records; Bilal gets 403 MODULE_ACCESS_DENIED and no Vault entry. After the switch-off Sara is refused on her next request.		
R-05	FR-G1	—	Upload a .exe renamed .pdf; a 12 MB photo; a 9 MB photo.	415 UNSUPPORTED_FILE (checked by the file's first bytes); 413 FILE_TOO_LARGE; the 9 MB photo is saved.		
R-06	FR-G1	Records added by Ali and by Sara.	Sara opens Ali's record and Ammi's record; then Ali opens Sara's record.	Sara: no Edit/Delete on others' records (can_modify false; a PATCH gives 403 RECORD_MODIFY_DENIED). Ali (Primary) can edit and delete any record.		
R-07	FR-G1	Phone network drops during an upload.	Retry the same upload (same client_id); then reuse that client_id for a different record.	Retry: 200 already_saved, only one record. Different record: 409 CLIENT_ID_REUSED.		
R-08	FR-G1	At least 6 records over 2 years, some with pictures and PDFs.	Export → Complete history → Make PDF.	Front page with name, period and a table of every record; pictures under their record; attached PDFs after the summary; a damaged PDF is listed as "Not included".		
R-09	FR-G1	Same.	Export One year (2026) and One month (Oct 2026); a month with no records; month without a year (Postman).	Only that period included; empty period still gives a clear PDF or message (record what happens); month without year: 422.		
R-10	FR-G1	A record with an Urdu title.	Export it.	Known limit: Urdu shows as ? inside the PDF (the app shows it correctly). Record as Low, not a failure.		
R-11	FR-G3	—	Make a doctor link (15f); open it in a private browser with no login; try to change anything.	201 with the link shown once; PDF opens read-only; Share / Copy work; the app adds the server address if the link starts with /.		
R-12	FR-G3	R-11 link.	Cancel the link and open it; make one with hours = 1 and open it after an hour; make 11 open links.	Cancelled: 410 SHARE_REVOKED; expired: 410 SHARE_EXPIRED; 11th link: 409 SHARE_LIMIT. Links list shows Active / Expired / Cancelled and open counts.		
R-13	FR-G3, A4	Sara made a link.	Ali switches Sara's health_records OFF; open Sara's link.	The link stops working.		
R-14	FR-N3	R-01 – R-12 done.	Ali opens the audit log (screen comes with Task 6.6; until then ask Abdullah to show the rows).	Every file read, export, add, change and delete is logged with who, what and when.		
R-15	FR-N2	—	Ask Abdullah to open a stored file directly on the server disk.	Unreadable (AES-256-GCM encrypted); the app still gets the original file.		
R-16	FR-G2	Task 5.2 built; details in A2.	Ali hides Prescriptions from Sara (15e); Sara opens the list, an export and a prescription by its id.	Prescriptions vanish from her list and exports; opening one by id gives 404 RECORD_NOT_FOUND (it looks like it does not exist).		

A2. Record categories per Secondary (FR-G2, Task 5.2)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
RA-01	FR-G2	Sara health_records ON; nothing hidden yet.	Sara opens Ammi's vault; GET …/records/access/me.	Sara sees all four categories (tabs and the category picker); access/me shows all four true. Nothing is hidden by default.		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
RA-02	FR-G2	Prescriptions and lab reports exist.	Ali → Record access (15e) → Sara → Prescriptions OFF. Sara refreshes the list, filters by Prescriptions, exports a PDF, opens a prescription and its file by id (Postman), tries edit / delete.	PUT 200; on her next request prescriptions are gone from her list, total and export; the filter gives an empty list (not an error); open / file / edit / delete give 404 RECORD_NOT_FOUND. The Prescriptions tab and picker choice are not shown at all.		
RA-03	FR-G2	RA-02 done.	Sara adds a record as Prescription; moves a lab report into Prescription; makes a doctor link with categories = prescription.	Each gives 403 CATEGORY_NOT_ALLOWED; nothing is saved.		
RA-04	FR-G2, G3	Before RA-02, Sara made two doctor links: one for all categories, one for prescriptions only.	After RA-02, open both links.	The all-categories link opens without the prescriptions; the prescriptions-only link gives 410 SHARE_REVOKED.		
RA-05	FR-G2	Sara added a prescription herself before RA-02.	Sara looks for it.	Hidden from her too (agreed rule, clarification 105). Record as expected behaviour.		
RA-06	FR-G2	RA-02 done; switch Bilal's health_records ON for this case.	Bilal, Ali and Ammi open the vault.	Bilal still sees prescriptions (the setting is per Secondary); Ali (Primary) and Ammi (Dependent) always see all four.		
RA-07	FR-G2, A4	Bilal health_records OFF.	Ali opens Record access (15e).	Bilal's four switches are greyed out with "Needs Health Records permission first" (health_records_granted false).		
RA-08	FR-G2	—	Postman: Sara calls GET …/records/access and a PUT; Ali sends a PUT for himself; a PUT with a guardian_id that is not a Secondary; a PUT for category "photos"; the same PUT twice.	403 NOT_PRIMARY_GUARDIAN; 400 PRIMARY_NOT_GOVERNABLE; 404 SECONDARY_NOT_FOUND; 422 UNKNOWN_CATEGORY; repeating the same PUT gives the same answer.		
RA-09	FR-G2, N3	RA-02 done.	Ali turns Prescriptions back ON for Sara; check the audit rows.	Prescriptions come back on Sara's next request; audit shows record_category_hidden and record_category_shown with who and when.		

B. Appointments (FR-G4)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
AP-01	FR-G4	Ali logged in.	Appointments → Add: Dr. Hamid, tomorrow at this time + 5 min.	201; shows under Upcoming (15d) and on Ammi's My health (28); Ammi gets an appointment_update notification, Ali (who added it) does not.		
AP-02	FR-G4	AP-01 saved.	Wait for the reminders (or ask Abdullah to set a test appointment ~1 h ahead).	Ammi, Ali and Sara (health_records ON) get appointment_reminder 24 h and 1 h before; Bilal does not.		
AP-03	FR-G4	—	Add one in the past; add the same doctor at the same time twice; send a time without a time zone (Postman).	422 APPOINTMENT_IN_PAST; 409 APPOINTMENT_EXISTS; 422 VALIDATION_ERROR.		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
AP-04	FR-G4	AP-01 saved.	Move it to another time; then cancel it; then try to change it.	Move: others notified, reminders follow the new time. Cancel: moves to History, others notified. Change after cancel: 409 APPOINTMENT_CANCELLED.		
AP-05	FR-G4	A cancelled and a past appointment.	Open History; delete one.	History shows both, latest first; delete removes it completely (204).		

C. Vitals (FR-H1)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
H-01	FR-H1	Ammi logged in.	My health (28) → Add my blood pressure → 138 / 88 → Save.	201; source manual, recorded by Ammi; Ali's Vitals (26) shows 138/88 mmHg with the time and "by Ammi".		
H-02	FR-H1	Ali logged in.	Add reading (26b): blood sugar 112 fasting; SpO2 97; temperature 36.9.	Each saved (201); tiles on 26 update; sugar shows "Fasting".		
H-03	FR-H1	—	Type sugar 6.2 mmol/L and temperature 98.4 °F.	Saved and shown only in mg/dL (about 111.7) and °C (about 36.9). The app never converts again.		
H-04	FR-H1	Postman or the app.	BP 400/20; BP 120/130 (bottom higher than top); BP with no bottom number; SpO2 120; temperature 50 °C; a diastolic on blood sugar; heart_rate typed by hand.	422 VITAL_OUT_OF_RANGE with details.field for the range errors; 422 VALIDATION_ERROR for the wrong fields; heart rate refused (band only).		
H-05	FR-H1	—	recorded_at in the future; 400 days ago; a time without a time zone.	422 for each (INVALID_DATE or VALIDATION_ERROR); nothing saved.		
H-06	FR-H1	Phone network drops while saving.	Retry with the same client_id; then reuse that client_id with other numbers.	Retry: 200 already_saved, one reading only. Other numbers: 409 CLIENT_ID_REUSED.		
H-07	FR-A4	Sara health_records ON, Bilal OFF.	Sara opens Vitals and adds a reading; Bilal opens Vitals; Zain (not linked) tries.	Sara can; Bilal gets 403 MODULE_ACCESS_DENIED (no Vitals entry); Zain 403.		
H-08	FR-H1	Readings by Ali and Sara.	Sara tries to delete Ali's reading; Ali deletes Sara's; look for Edit.	Sara: 403 VITAL_MODIFY_DENIED (no Delete button, can_delete false). Ali (Primary) can. No Edit: delete and add again.		
H-09	FR-H1	Readings on 7 different days.	Vitals → 7 days, then 30 days; filter one metric.	Chart points match readings and dates; latest tile per metric is the newest; heart rate is empty until the band sends data.		
H-10	Safety	—	Look through 26, 28, 16 and the API replies for "high", "low", "normal" or coloured warnings.	None: numbers only, no medical advice (contract rule). The app may say "ask the doctor".		
H-11	FR-H1	Readings + band data exist.	Export vitals PDF (complete / year / month); records export with Add vitals ticked (15c); a doctor link with vitals.	Separate vitals PDF works; the vault PDF and the doctor's PDF get a Vitals section only when ticked; band readings appear as one line per day (average, lowest, highest, count).		
H-12	FR-N3	H-01 – H-11 done.	Ask Abdullah to show the audit rows (or the audit screen once built).	Add, delete and export of vitals are logged with who and when.		

D. Wearable band — mock data (FR-H2, NFR-13)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
W-01	FR-H2	Ammi logged in.	Connect band (27b) → pick Band 7A2F; pair the same band again; pair a different band; Sara tries to pair.	201 paired; same band again 200; different band 409 BAND_ALREADY_PAIRED (unpair first); Sara 403.		
W-02	FR-H2	Just paired, no data yet.	Open My band (27) and Ali's Vitals (26).	Status waiting_for_first_data; heart rate empty; no alert.		
W-03	FR-H2	Paired.	Make demo data (POST …/wearable/mock, 30 readings).	Status receiving; heart rate shows on 27 and on the 26 tile as "Band · x min ago"; readings have source wearable. No battery in the app (3 LEDs on the band).		
W-04	FR-H2	Postman as Ammi.	Sync a batch with one good reading, the same reading twice, heart rate 300, a reading 1 hour in the future, one 8 days old.	200; only the good one stored; duplicates 1; rejected lists the other three by index. Nothing else is blocked.		
W-05	FR-H2	Paired and receiving; Sara health_records ON, Bilal OFF.	Stop data for 30 minutes (ask Abdullah to shorten the test time if possible).	Ali and Sara get a STANDARD band_no_data notice (26c); Bilal does not; it never climbs the escalation chain. New data → all-clear (resolved: true).		
W-06	FR-H2	W-05 set-up.	Upload only old readings stored on the band; separately, take Ammi's phone offline.	Old readings do not close or open the notice; phone offline gives the device_offline alert, not band_no_data.		
W-07	NFR-13	Band silent (W-05).	Ammi presses Panic, confirms a dose, sends a message.	All work normally; a silent band blocks nothing.		
W-08	FR-H2	Paired.	Ali calls the sync endpoint (Postman); Ammi unpairs (Disconnect band) twice.	Ali 403 (only the Dependent's phone syncs). Unpair 204, safe to repeat; old readings stay; an open no-data notice closes.		
W-09	NFR-13	Real band (Task 5.5).	Walk out of range, come back.	Reconnects by itself within 30 s. BLOCKED until 5.5.		
W-10	Security	Real band + a second band nearby (Task 5.5).	Try to pair with a band that is not Ammi's.	Only the paired band's data is accepted. BLOCKED until 5.5.		

3. Open points

- Screen 15e lists each Secondary by name, but GET …/records/access (like the permissions endpoint) gives only email and phone; until a name is added the app shows the email.
- Empty export period (R-09): the contract does not say what happens; record the behaviour and log it as a question if unclear.
- Band no-data time is 30 minutes (W-05); a shorter test setting on the server would make the run faster (ask Abdullah only when the run is scheduled).
- Band: custom ESP32 build; W-09 and W-10 wait for the real band (Task 5.5). Steps are not sent yet (sensor list open); battery only by the 3 LEDs.